AP PRECALCULUS

Unit 1 • Worksheet 1.2D

Worksheet 1.2D • Rates of Change

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Using the interval $[2, 2.001]$, which is the best estimate of the rate of change of $f(x) = x^2 + x$ at $x = 2$?

(A) 4
(B) 6
(C) 2.5
(D) 5

Question 2

NO CALCULATOR

The function g changes from increasing to decreasing at $x = 3$. Which must be true about the rate of change of g at $x = 3$?

(A) It changes from negative to positive.
(B) It changes from positive to negative.
(C) It is undefined.
(D) It is the greatest value of g.

Question 3

CALCULATOR

Which is the best estimate of the rate of change of $f(x) = \sqrt{x}$ at $x = 4$?

(A) 0.125
(B) 0.25
(C) 0.5
(D) 2

Question 4 (Free response)

NO CALCULATOR

A ball is thrown upward. Its height is $h(t) = -5t^2 + 20t + 2$ meters, t seconds after it is thrown.

(a) Find the average velocity of the ball on $[1, 2]$, $[1, 1.1]$ and $[1, 1.01]$.
(b) Use (a) to estimate the velocity of the ball at $t = 1$, with units.
(c) Estimate the velocity at $t = 3$ using $[2.99, 3]$ and $[3, 3.01]$. What does its sign mean?

Question 5 (Free response) NO CALCULATOR

Explain, using rates of change, why the graph of a function that is increasing and concave down rises more and more slowly.